

CFD Midterm

By: Fourie van Rooyen, Harrison Davis, and Tony Holeman

1) 3D Rayleigh Bernard Convection

Governing Equations + Major Assumptions:

Gravity: $g = -g\vec{z}$

Incompressible Continuity: $\nabla \cdot \vec{u} = 0$

Navier Stokes with Buoyancy: $\rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) = -\nabla P + \mu \nabla^2 \vec{u} + \rho_0 \beta (T - T_0)(-\gamma \vec{z})$

Energy Equation: $\rho_0 C_p \left(\frac{\partial T}{\partial t} + \vec{u} \cdot \nabla T \right) = k \nabla^2 T$

Boussinesq Assumption: Density variation ignored except those resulting from gravity

Incompressible: Constant $\rho_0, \mu, K, \text{ and } C_p$

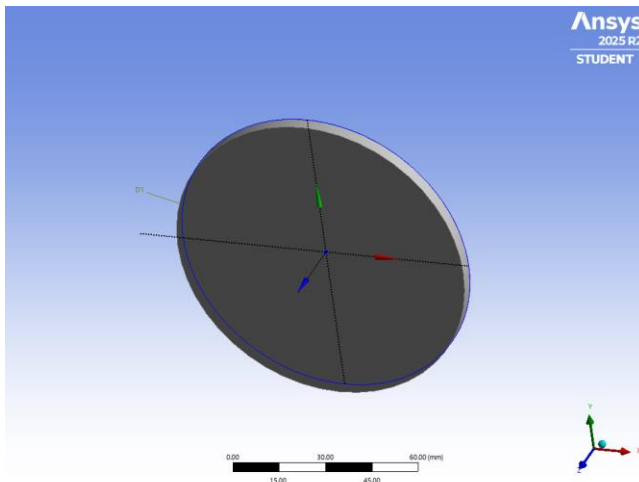
Laminar, Viscous Flow: No turbulent modeling

Isothermal, No-Slip Walls: $\vec{u} = 0$ at walls with fixed temperatures

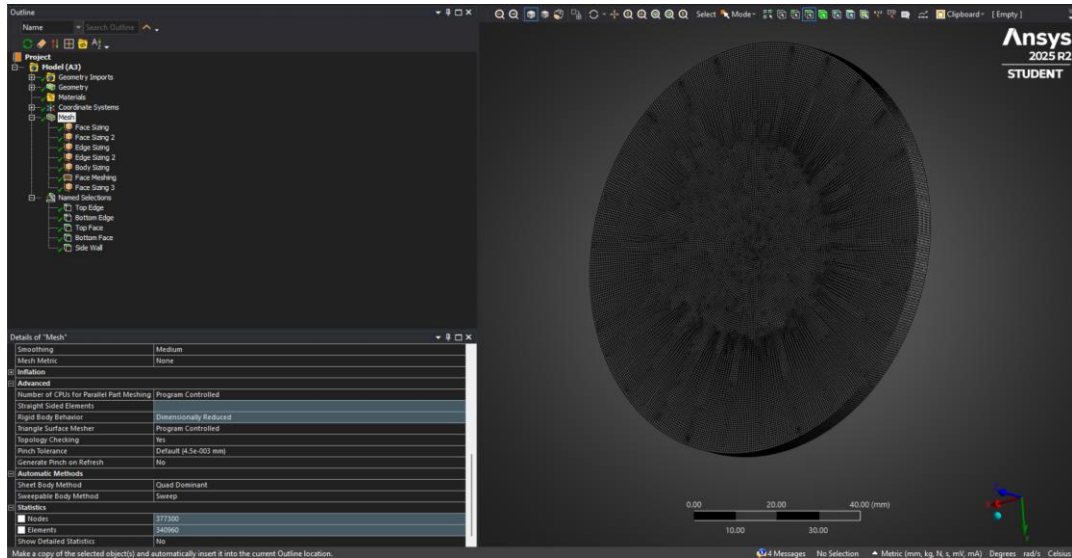
No Internal Heat Generation: No volumetric heat in energy equation

No Viscous Dissipation: Negligible, leading to no viscous heat in energy equation

Geometry:



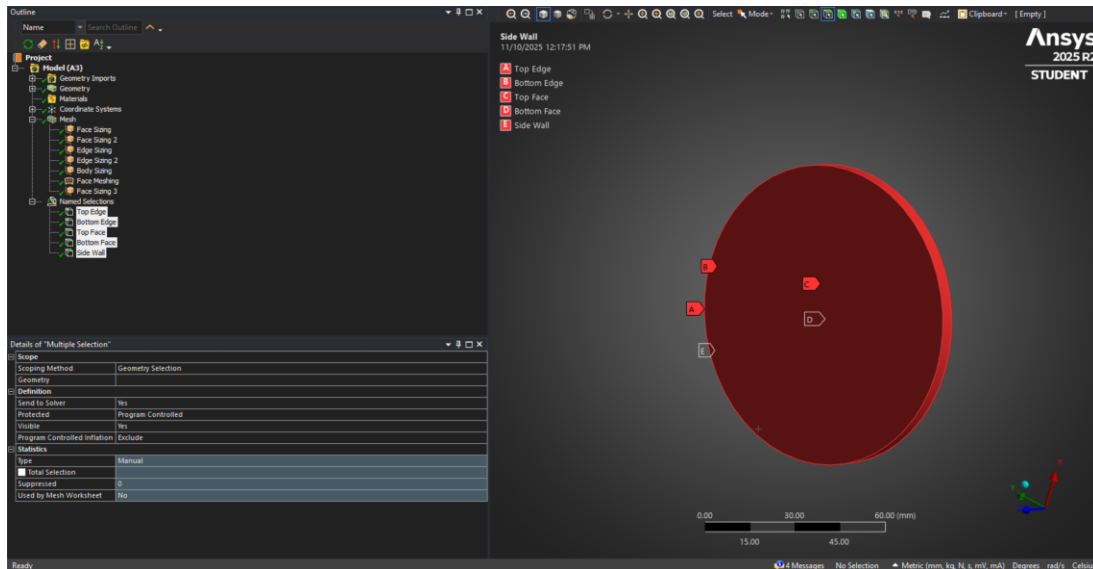
Mesh:



From image above:

Nodes = 377,300 Elements = 340,960

Named Selection:

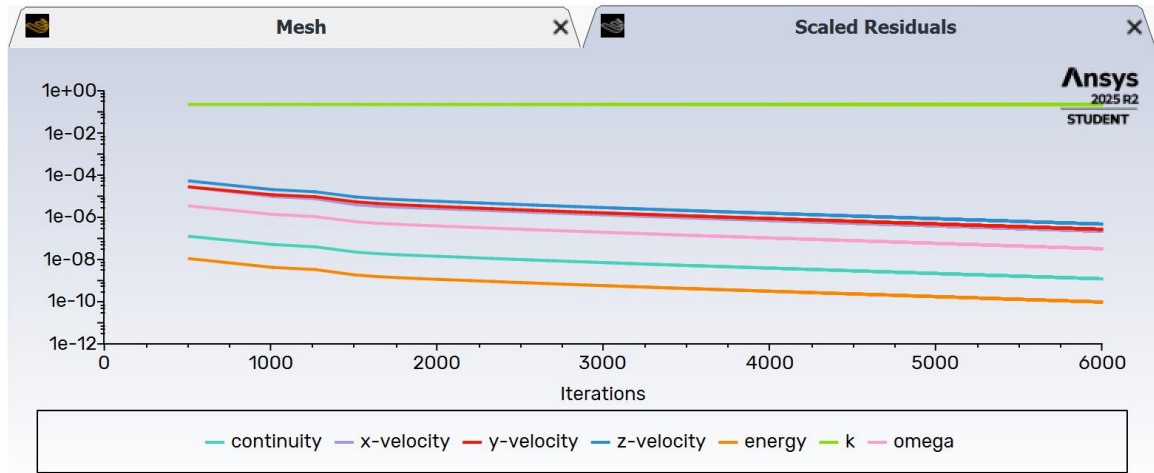


Physics Setup:

- 1) Yes, we need the energy equation.
- 2) Boussinesq Model

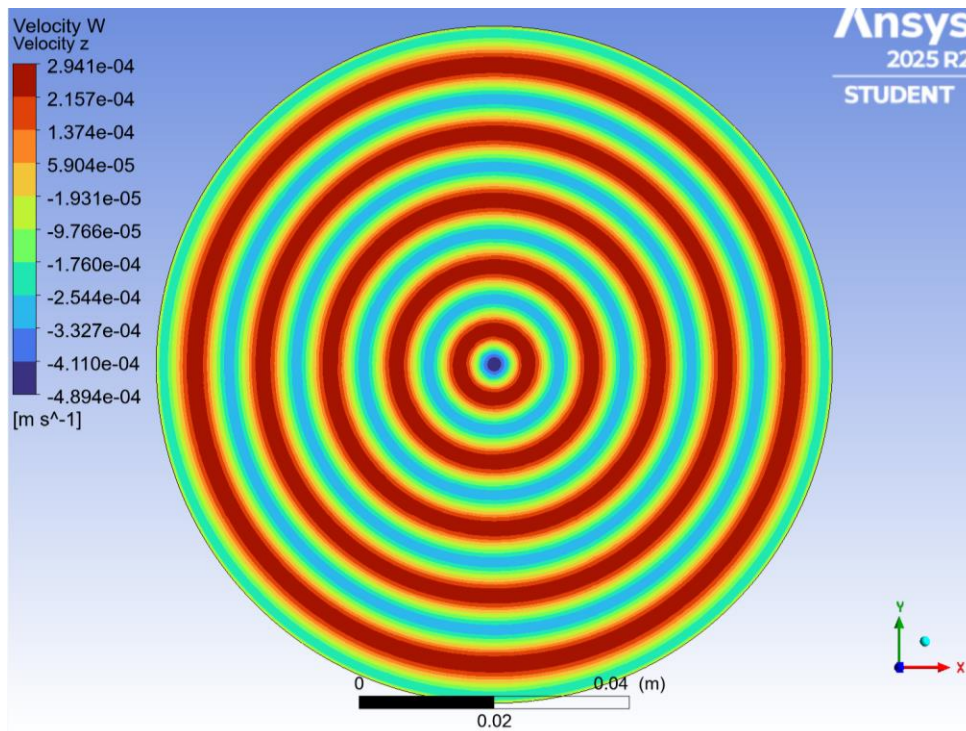
$$3) R_a = \frac{g\alpha\Delta TL^3\rho^2c_p}{\mu k} = \frac{(9.81)(0.000247)(298-296.7)(0.0048)^3(997)^2(4180)}{(0.000891)(0.607)} = 2676.3$$

4) Residual Plots Showing Convergence:

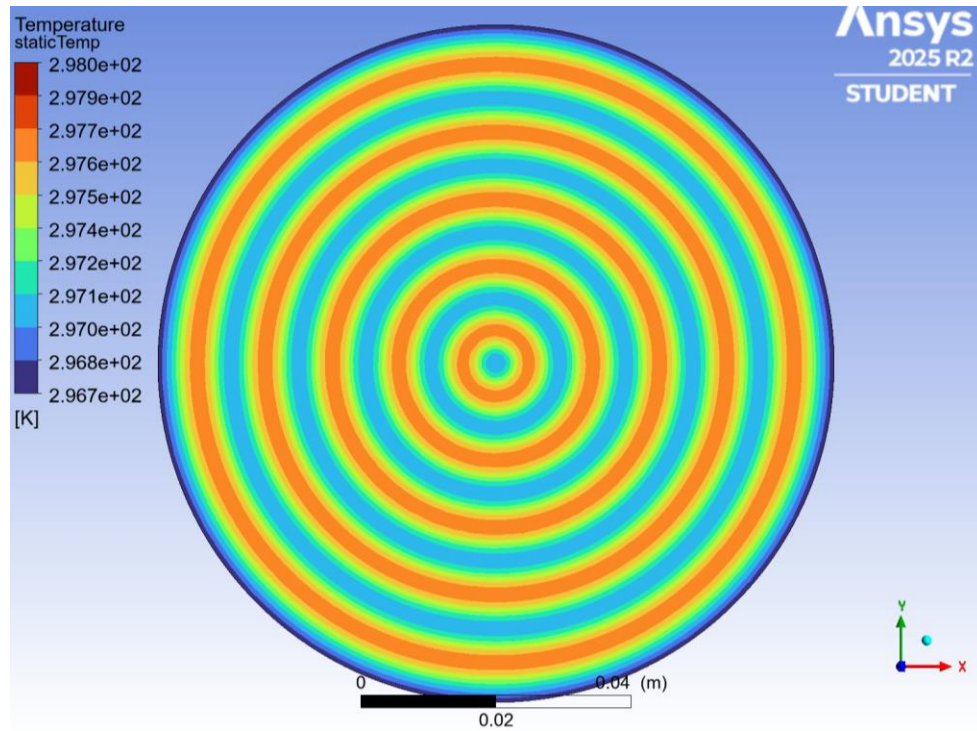


Results:

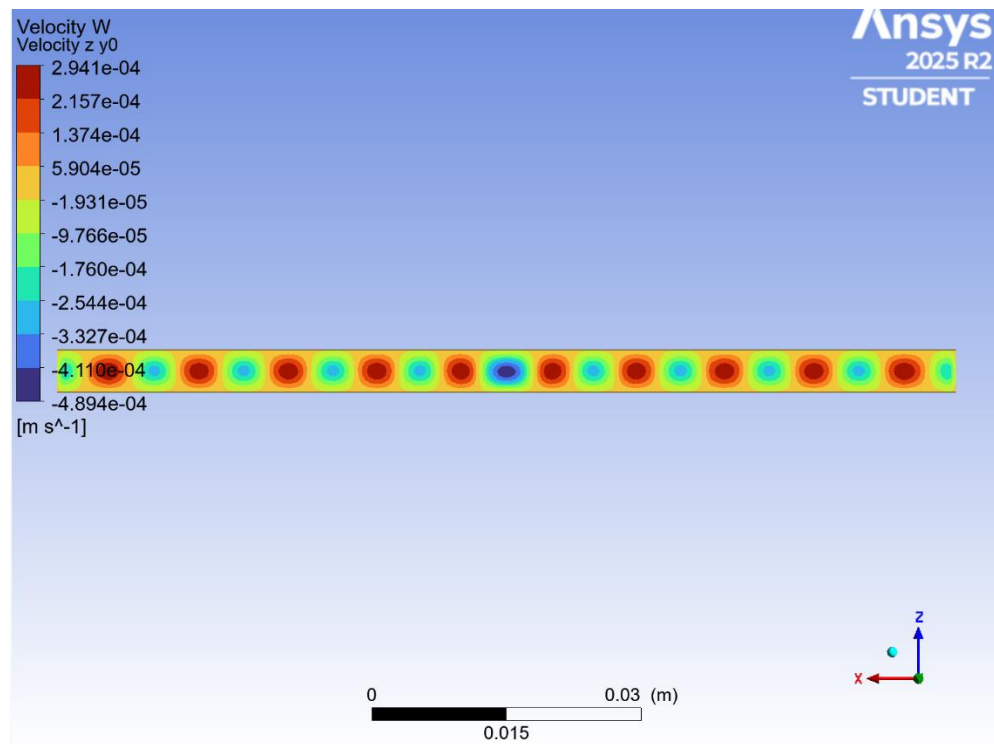
Z Velocity Plot at Midplane:



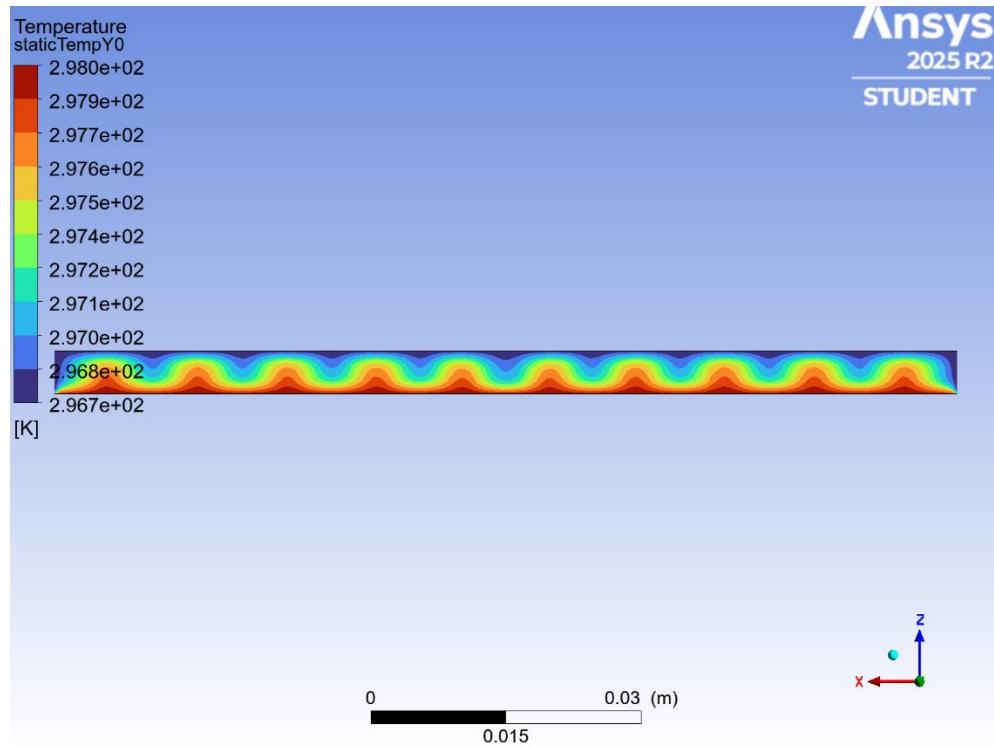
Static Temperature Plot at Midplane:



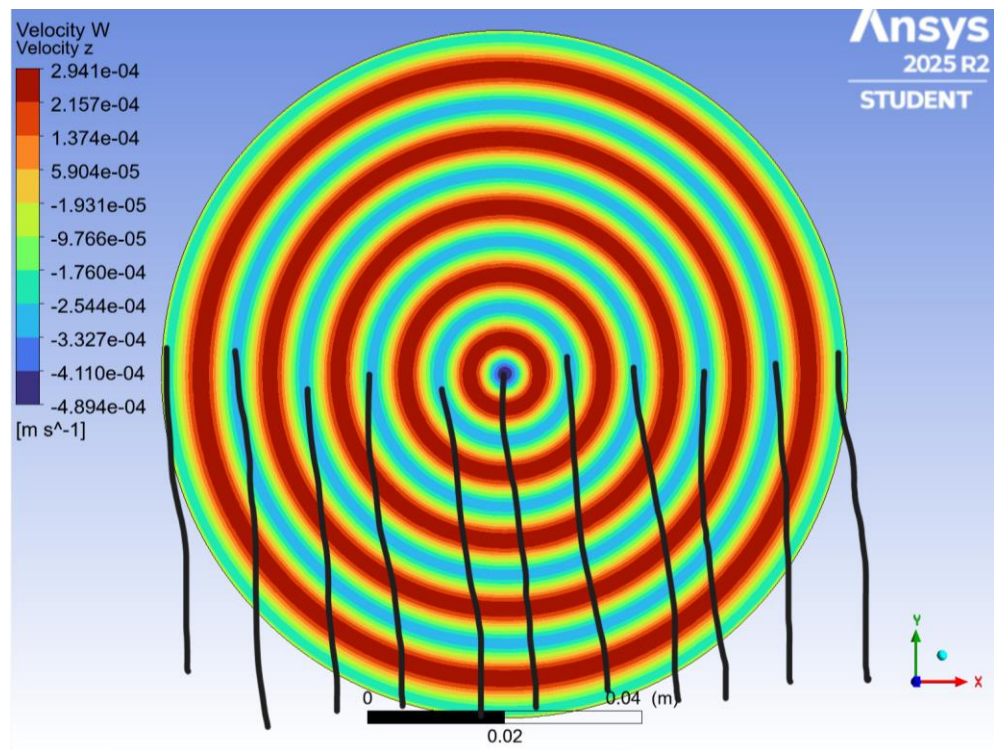
Z Velocity Plot at XZ Plane:



Static Temperature Plot at XZ Plane:



Extra Credit:



From left to right, velocity z drops to blue region 11 times. Thus, the total waves amount equals number of repeating regions between blue sections, thus $n = 11 - 1 = 10$ waves.

$$\gamma = \frac{L}{n} = \frac{0.1m}{10} = 0.01m$$

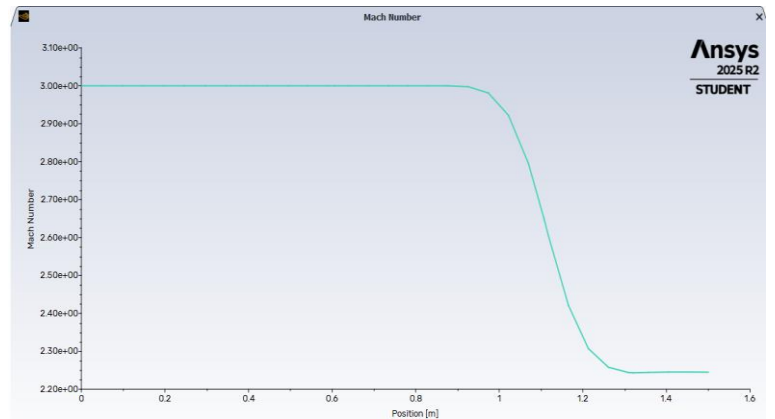
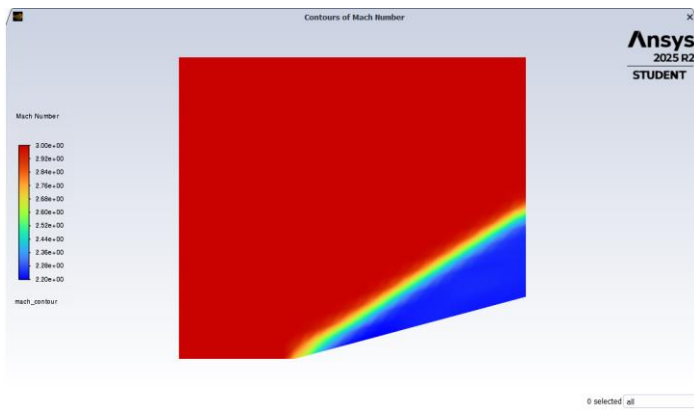
$$k_{crit} = 2\pi \frac{H}{\gamma} = 2\pi \frac{0.0048m}{0.01m} \approx 3.016$$

$$\text{Percent Error} = \left| \frac{3.016 - 3.2}{3.2} \right| \cdot 100\% \approx 5.75\% \text{ Error}$$

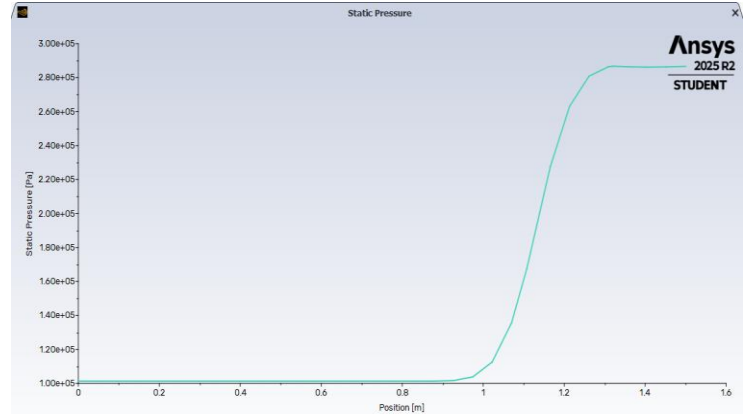
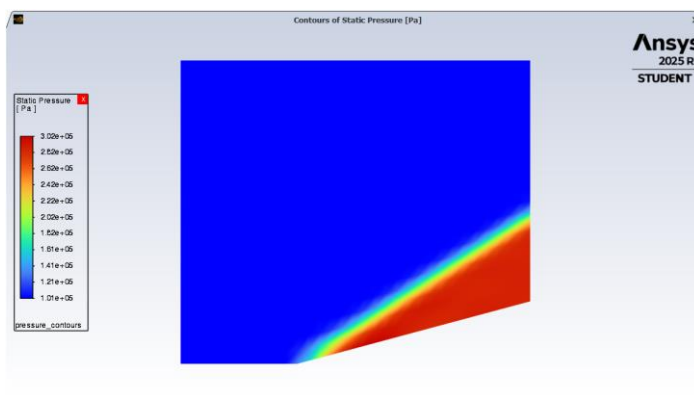
2) Inviscid Supersonic Flow w/ Wedge

A) *Cone vs Wedge:*

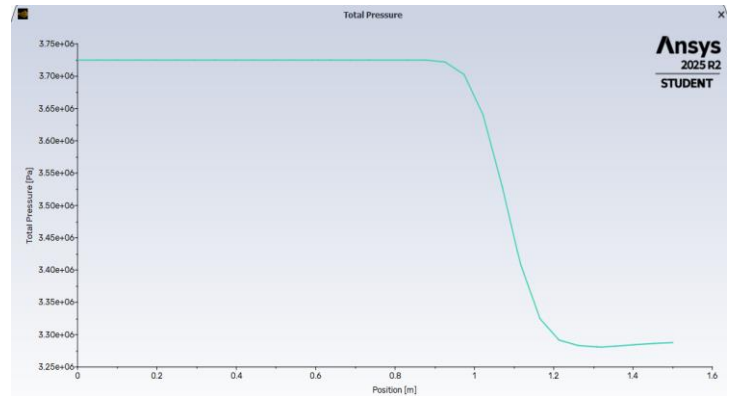
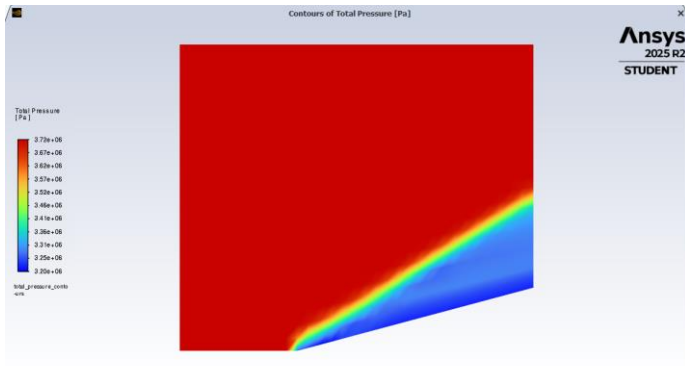
Wedge Mach Number:



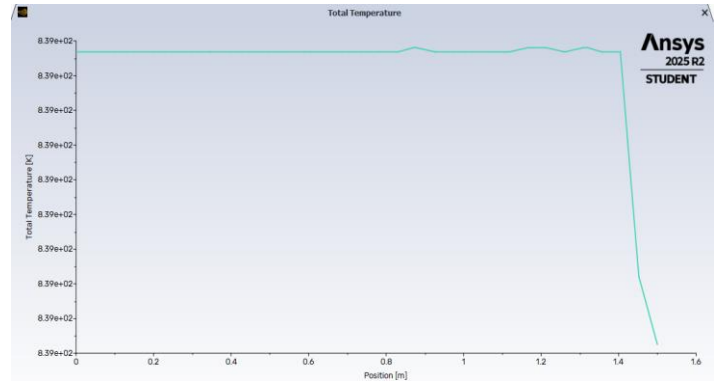
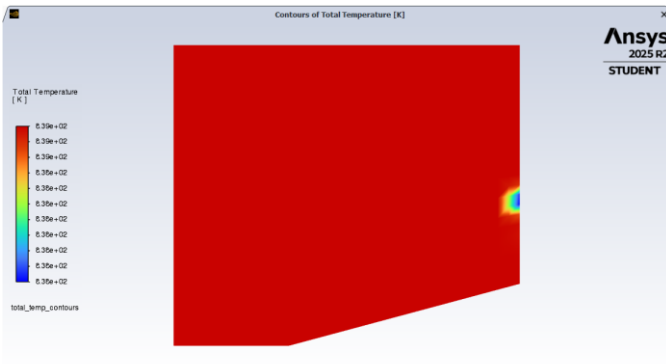
Wedge Static Pressure:



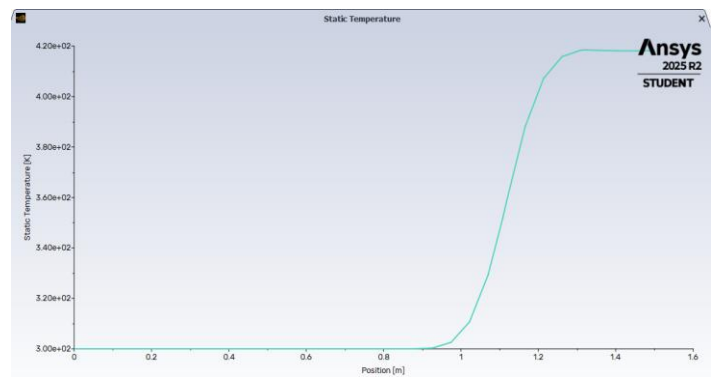
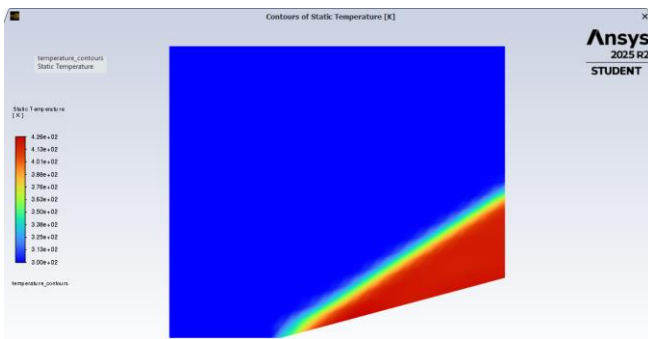
Wedge Total Pressure (Pa):



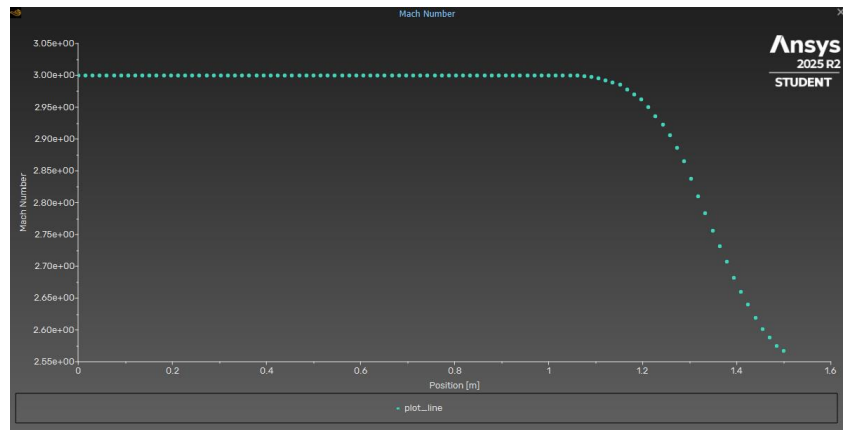
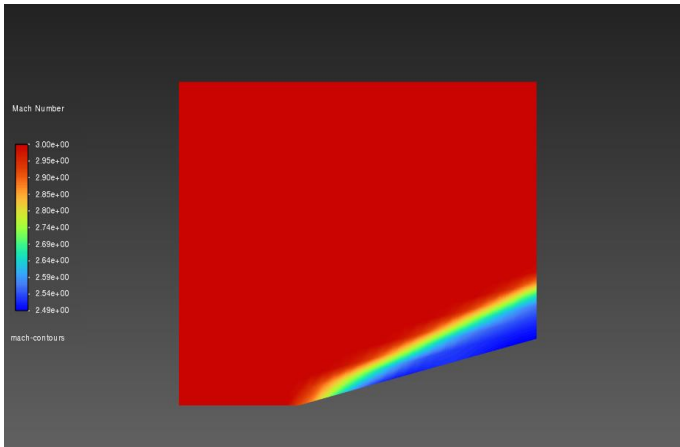
Wedge Total Temperature (K):



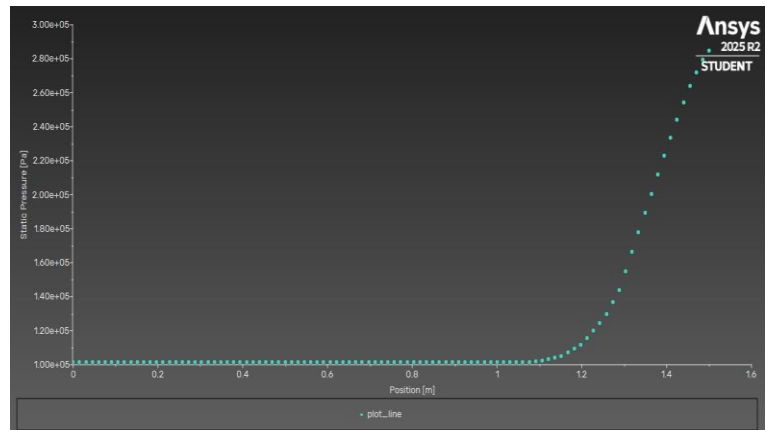
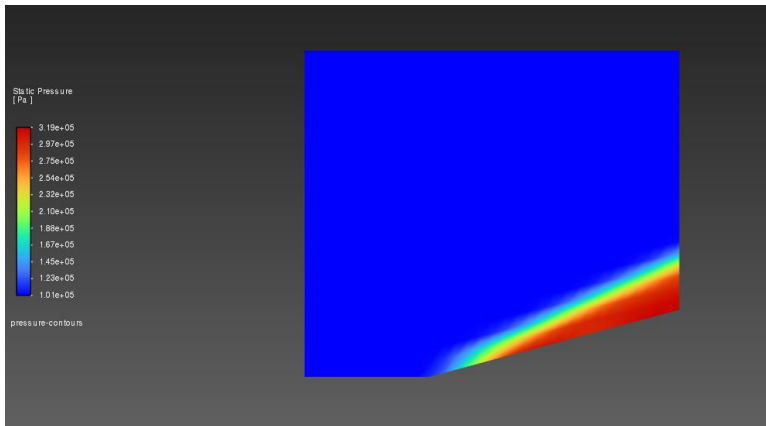
Wedge Static Temperature:



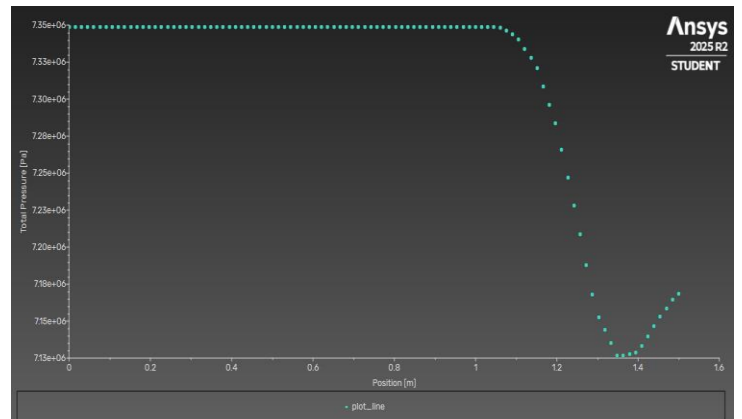
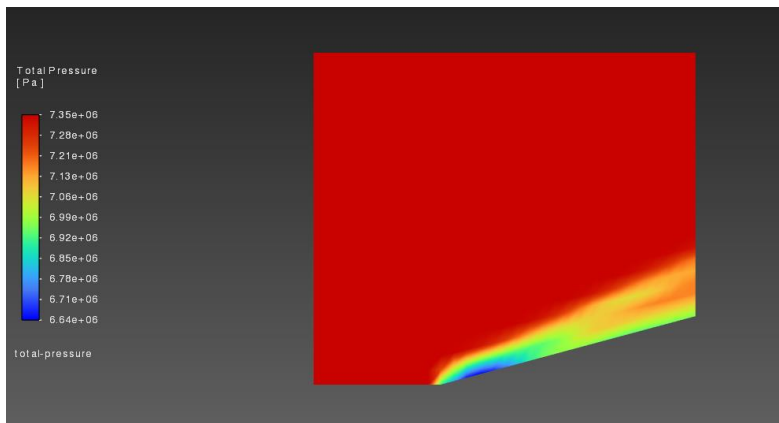
Cone Mach Number:



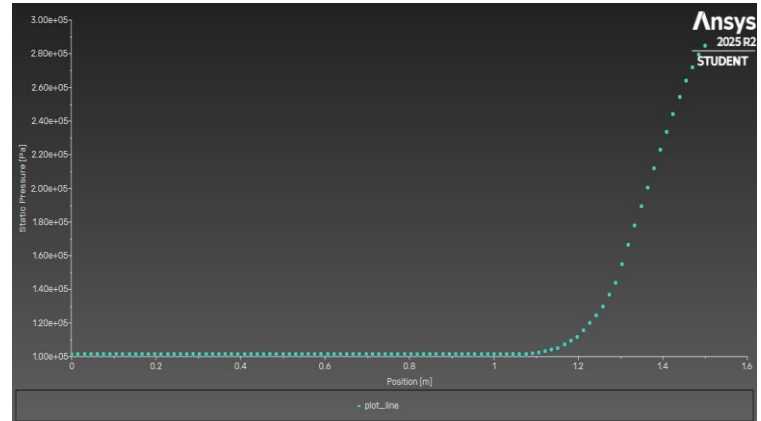
Cone Static Pressure:



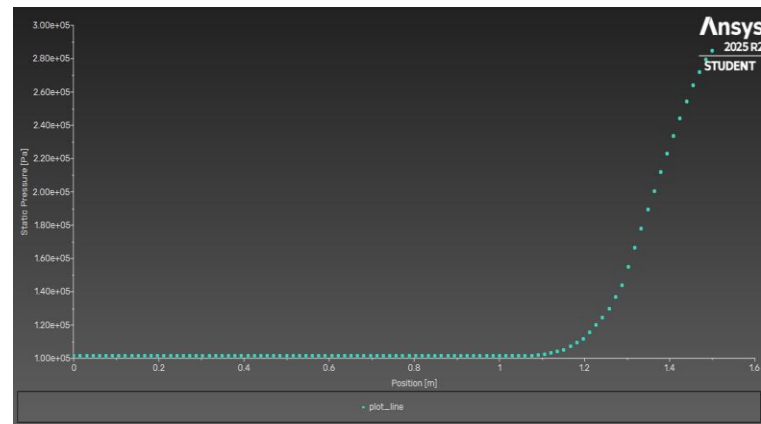
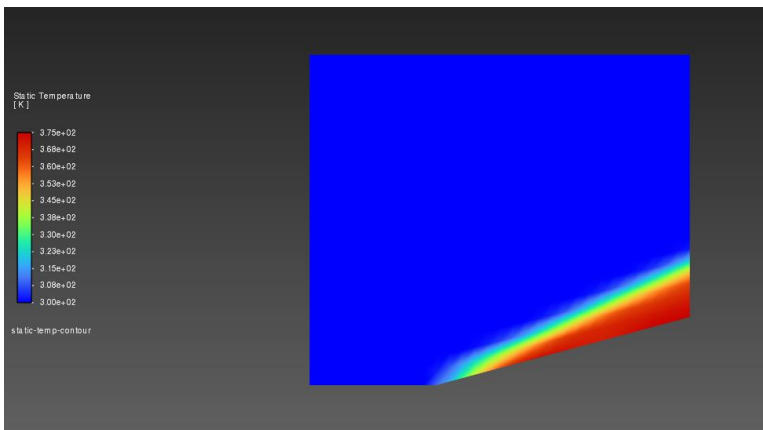
Cone Total Pressure:



Cone Total Temperature:



Cone Static Temperature:



Comments:

In the wedge results, an oblique shock forms at the wedge, indicated by distinct pressure and temperature gradients. The shock's effect appears in all contours except total temperature, which remains uniform aside from a minor simulation error. The XY plots further illustrate this behavior; since the y-line is set at 0.4, the shock location corresponds to the sharp change in values around $x = 1$. The total temperature plot does not show this trend due to its uniformity, and the sharp drop near the end reflects the same simulation error.

The axisymmetric and wedge cases produce notably different results. The shock in the axisymmetric case is slightly curved and forms at a shallower angle than in the wedge simulation. Across all pressure and temperature contours (except total temperature), the axisymmetric shock shows smaller changes in magnitude. This is expected, as the flow expansion around the cone's circumference weakens the shock, while the wedge's restricted flow produces a much stronger one.

B) Free Stream Mach Cases:

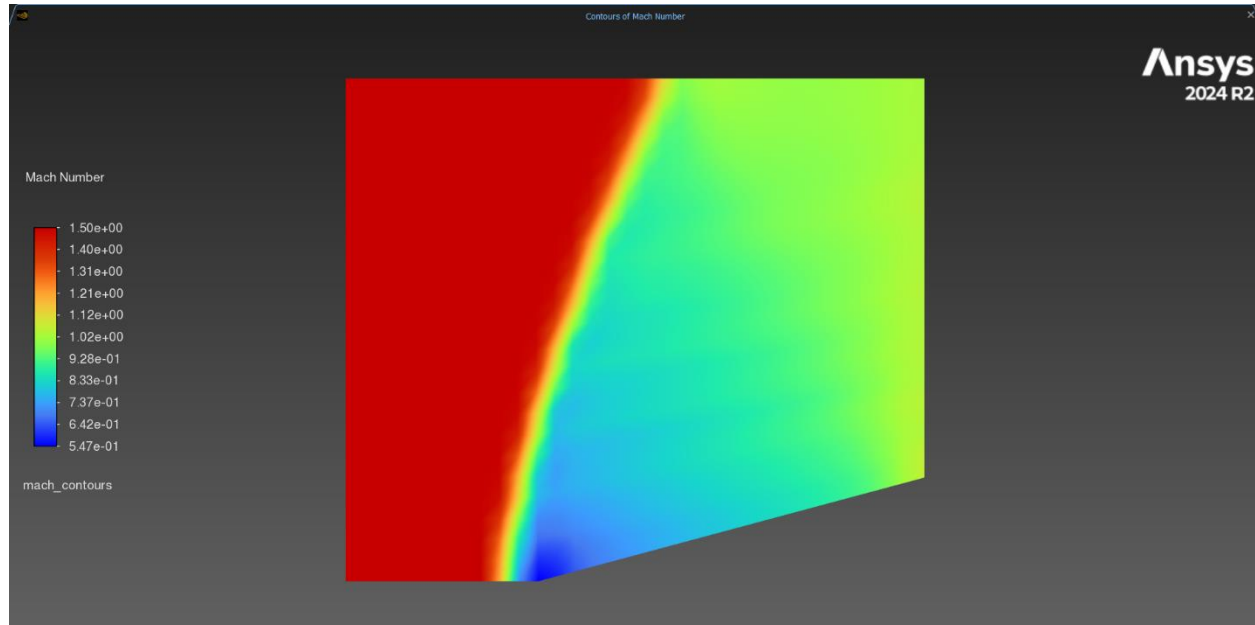
Comments:

For a 15° wedge, oblique shock theory says you can only get an attached shock if the free-stream Mach number is above around 1.6-1.8. So, at $M_\infty = 2.0$ and $M_\infty = 1.8$, the flow can turn over the wedge and an attached oblique shock forms at the tip which means that the simulation should converge fine. But at $M_\infty = 1.5$, the Mach number is too low for an attached oblique shock at the tip because the flow cannot turn enough and the shock detaches which forms a bow shock in front of the wedge. This matches what theory predicts. For Mach numbers below around 1.6-1.8, a 15° wedge can't have an attached shock.

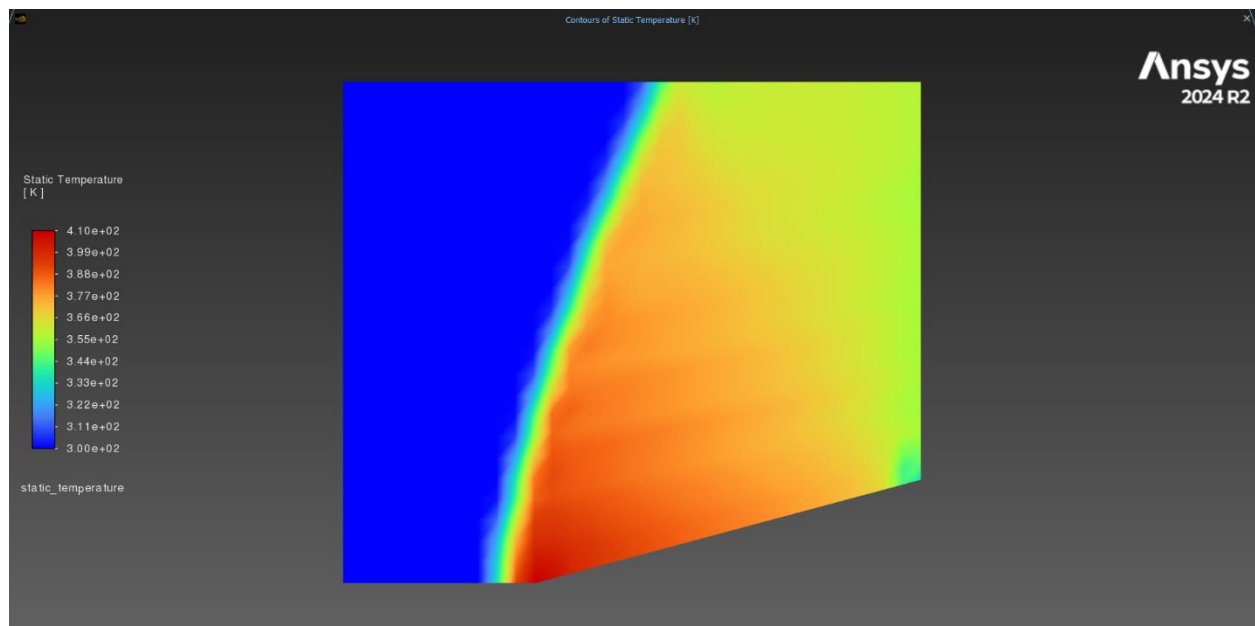
(1.5 Free Stream Mach Contours on Next Page)

M_∞ 1.5 Contours:

Mach Number:



Static Temperature:



Static Pressure:

